

Education

Department of Biological Sciences, University of Pittsburgh

PH.D. CANDIDATE IN MOLECULAR, CELLULAR, AND DEVELOPMENTAL BIOLOGY

Pittsburgh, PA, USA

Aug 2021 – Expected Apr 2027

- Advisor: **Dr. Jacob Durrant**
- Research focus: cheminformatics, structure-based drug design, and scientific software development.

University of Science and Technology, Zewail City

B.S. IN COMPUTATIONAL BIOLOGY AND GENOMICS (BIOMEDICAL SCIENCE)

Giza, Egypt

Aug 2017 – Jul 2021

- Advisor: **Dr. Eman Badr**
- Graduated *cum laude*; awarded a **90% merit-based scholarship** throughout all four years of study.

Research Experience

Graduate Student Researcher

Pittsburgh, PA, USA

DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF PITTSBURGH

Sep 2021 – Apr 2027

- **LIGNOVA: large-scale protein–ligand complex generation pipeline:** Designed and built a fully automated, open-source pipeline that docks bioactive compounds against experimentally determined protein structures, yielding over **232 million docked poses** (240,124 ligands across 10,851 protein structures; 5.27M protein–ligand pairs) to date — assembling large-scale receptor-bound structural data for structure-based drug design and ML training.
- **reqadence: open-source async API-client library:** Developed an open-source, fully-typed asynchronous Python package (automatic retries, rate limiting, RFC-9111 response caching) that powers LIGNOVA's large-scale data ingestion from RCSB and PubChem.
- **Water modeling in protein–ligand docking:** Investigating whether **consensus-based rescoring** can substitute for explicit water modeling in docking by recovering the pose-prediction accuracy that ordered water provides while preserving the throughput required for large-scale virtual screening. Benchmarking curated high-quality X-ray complexes across multiple hydration conditions to characterize when consensus scoring can recover near-native poses in the absence of water. **Manuscript in preparation.**
- **Structure-based discovery of arginine sensors in yeast** (collaboration with the O'Donnell Lab): Identified **4 candidate arginine-sensing proteins** in *S. cerevisiae* invisible to sequence-homology approaches, combining Foldseek structural search against 5 human CASTOR1 structures with MolModa docking. Experimental validation confirmed altered arginine-homeostasis phenotypes in the identified candidates.
- Developed and maintained Python pipelines with **HPC automation** for large-scale structural and molecular data processing across all projects.

Undergraduate Student Researcher

Giza, Egypt

UNIVERSITY OF SCIENCE AND TECHNOLOGY, ZEWAIL CITY

Jun 2020 – Apr 2021

- Trained Random Forest regression models (scikit-learn) to predict **drug sensitivity from transcriptomic profiles** in acute myeloid leukemia patients, identifying genetic markers correlated with differential drug response.

Publications

Holland, M.*; **Ahmed, M.***; Young, J. M.; Drurey, J. R.; McFadyen, S.; Ostrowski, E. A.; Levin, T. C. Hypermutable hotspot enables the rapid evolution of self/non-self recognition genes in *Dictyostelium*. *PNAS* **2025**, 122(51). DOI: 10.1073/pnas.2520843122. (*co-first authors)

Kochnev, Y.; **Ahmed, M.**; Maldonado, A. M.; Durrant, J. D. MolModa: Accessible and secure molecular docking in a web browser. *Nucleic Acids Research* **2024**, 52(W1), W498–W506. DOI: 10.1093/nar/gkae406.

Ahmed, M.; Maldonado, A. M.; Durrant, J. D. From byte to bench to bedside: Molecular dynamics simulations and drug discovery. *BMC Biology* **2023**, 21(1), 299. DOI: 10.1186/s12915-023-01791-z.

Honors & Awards

2026	NSF ACCESS Allocation (BIO260240) , Competitive national allocation on the Purdue Anvil and TAMU ACES clusters to scale the LIGNOVA pipeline	USA
2026	GPSG Travel Grant , Graduate & Professional Student Government, University of Pittsburgh	Pittsburgh, PA
2024 & 2026	Elizabeth Baranger Excellence in Teaching Award (Nominated) , A&S Graduate Student Organization, University of Pittsburgh; nominated by undergraduate students for teaching Biochemistry	Pittsburgh, PA
2021	Provost's Honors Roll , Zewail City of Science and Technology; awarded to the top 5 students by cumulative GPA	Giza, Egypt
2021	Cum Laude , B.S. Computational Biology & Genomics, Zewail City of Science and Technology	Giza, Egypt
2017–2021	Merit-Based Scholarship (90%) , Full merit scholarship for undergraduate studies, Zewail City of Science and Technology	Giza, Egypt

Conferences & Talks

2026	Poster Presenter — LIGNOVA: An Automated Pipeline for Large-Scale Generation of High-Quality Protein–Ligand Complexes, Computational Medicinal Chemistry School	Cambridge, MA
2026	Speaker — Putting the Pieces Together <i>in silico</i> to Build Drugs from Fragments, MCDB Noon Seminar Series, University of Pittsburgh	Pittsburgh, PA
2024	Speaker — DeepFrag Meets LIGNOVA: Transforming Drug Design with Fragment-Based Lead Optimization, MCDB Noon Seminar Series, University of Pittsburgh	Pittsburgh, PA
2024	Poster Presenter — Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, Gordon Research Conference & Seminar on Computational Chemistry	Portland, ME
2024	Speaker — Beyond the Blueprint: A Novel Database for Innovative Drug Design and Discovery, MCDB Noon Seminar Series, University of Pittsburgh	Pittsburgh, PA
2023	Poster Presenter — Augmenting Protein–Ligand Complex Databases with PubChem for Enhanced Drug Discovery, Gordon Research Conference & Seminar on Computer-Aided Drug Design	West Dover, VA
2022	Speaker — Testing the Impact of a Newly Built Receptor–Ligand Dataset on DeepFrag Accuracy, MCDB Noon Seminar Series, University of Pittsburgh	Pittsburgh, PA

Skills

Programming	Python, Bash, R, MATLAB; object-oriented design
Computational Tools	Git, GitHub, Linux/SLURM, HPC cluster management
Machine Learning	Supervised Learning, Feature Engineering, Model Evaluation, Graph Neural Networks, Random Forest
Data Analysis	Statistical Modeling, Data Visualization (Matplotlib, Seaborn)
Packages	PyTorch, scikit-learn, NumPy, pandas, Polars, SciPy, RDKit, Biopython, MDAnalysis, PyArrow
Cheminformatics	Docking (Glide, GNINA), Structural Search (Foldseek), Preparation (PDB2PQR, Gypsum-DL, Open Babel), Visualization (UCSF ChimeraX)
Genomics	Genome assembly (Flye, Nanopore/Illumina), alignment (minimap2, BWA, samtools), variant calling (bcftools, bedtools); BLAST, BUSCO, MScanX, CRAQ; FastQC
Languages	English (Fluent), Arabic (Native)

Teaching & Mentorship Experience

Undergraduate Research Mentor	Pittsburgh, PA, USA
TECBIO REU, DEPARTMENT OF COMPUTATIONAL & SYSTEMS BIOLOGY, UNIVERSITY OF PITTSBURGH	May 2024 – Jul 2024
- Co-mentored an undergraduate student on a 10-week project applying AlphaFold2 to explore HIV protease conformations for drug design. - Supervised development of a research poster, presented at the Research Symposium at Duquesne University.	
Graduate Teaching Assistant & Fellow (Biochemistry)	Pittsburgh, PA, USA
DEPARTMENT OF BIOLOGICAL SCIENCES, UNIVERSITY OF PITTSBURGH	Spring 2024, Fall 2025
- Conducted recitation sections and held weekly office hours for ~200 students; mentored 15 undergraduate teaching assistants. - Nominated for the Elizabeth Baranger Excellence in Teaching Award based on outstanding student evaluations.	
Undergraduate Teaching Assistant (Python)	Giza, Egypt
UNIVERSITY OF SCIENCE AND TECHNOLOGY, ZEWAIL CITY	Sep 2020 – Jan 2021
Guided students in writing and debugging Python code; held weekly office hours.	

Leadership & Service

Pitt Biological Sciences Graduate Student Organization	Pittsburgh, PA, USA
ARTS & SCIENCES GRADUATE STUDENT ORGANIZATION REPRESENTATIVE	Aug 2024 – Sep 2026
- Represent Biological Sciences at school-wide GSO meetings, addressing department concerns. - Coordinate with the GSO executive board on event planning, policy discussions, and decision-making processes. Served on the Summer Research Grant panel, evaluating graduate student research proposals across different A&S departments.	
Pitt Biological Sciences Graduate Student Organization	Pittsburgh, PA, USA
GRADUATE DIVERSITY, INCLUSION, AND BELONGING (DIB) COMMITTEE MEMBER	Sep 2023 – Dec 2024
- DIB works on fostering an inclusive environment for underrepresented students in the department. - Organize intercultural celebrations and distribute informational resources highlighting the diverse cultural traditions within the department.	

Zewail City Opencourseware

Giza, Egypt

VICE PRESIDENT

Jan 2018 – Mar 2020

- Led a team making university course materials (lecture notes, videos, assignments) freely available online for Egyptian students.
- Selected courses to document each semester, coordinated sub-teams, and contributed LaTeX-typeset lecture notes and solutions.

Bio Team Egypt

Egypt

HEAD OF STRATEGY COMMITTEE

Jan 2017 – Sep 2017

- Managed 3 subcommittees (Events, Awareness Campaigns, Charities) and organized 3 biotechnology seminars.
- Coordinated 4 public awareness campaigns introducing biotechnology to high school students across Egypt.